

Question 1) There are several ways to prove the Pythagorean theorem.

Consider triangles A and B.

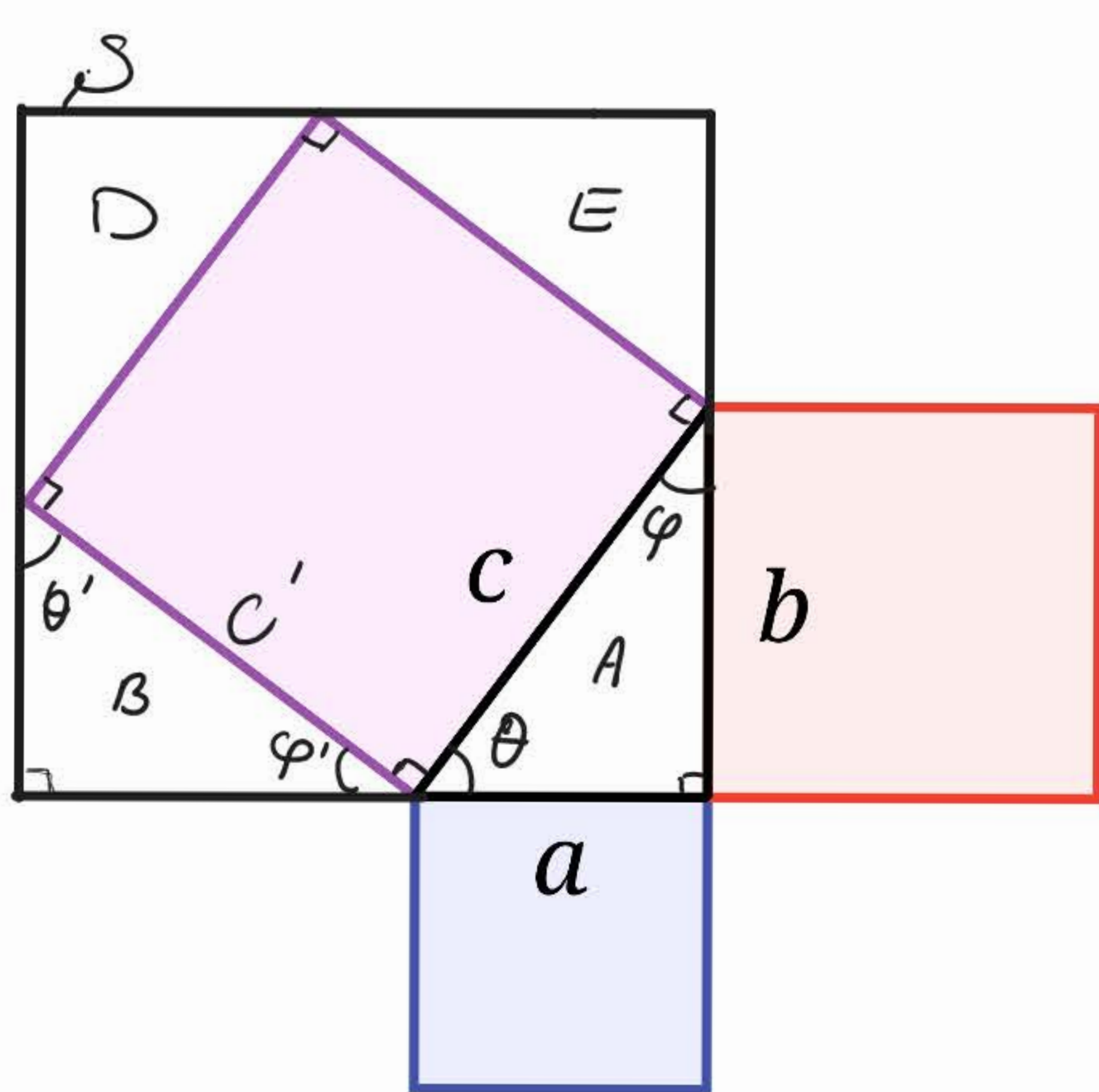
From the Euclidean geometry axioms, it follows that $\theta' = \theta$ and $\varphi' = \varphi$. Given $c = c'$, it follows that B and A triangles must be identical.

The area of the square S is:

$$\text{area of } S = (a+b)^2 = a^2 + b^2 + 2ab$$

$$= c^2 + 4\left(\frac{a+b}{2}\right)$$

$$\Rightarrow a^2 + b^2 = c^2$$

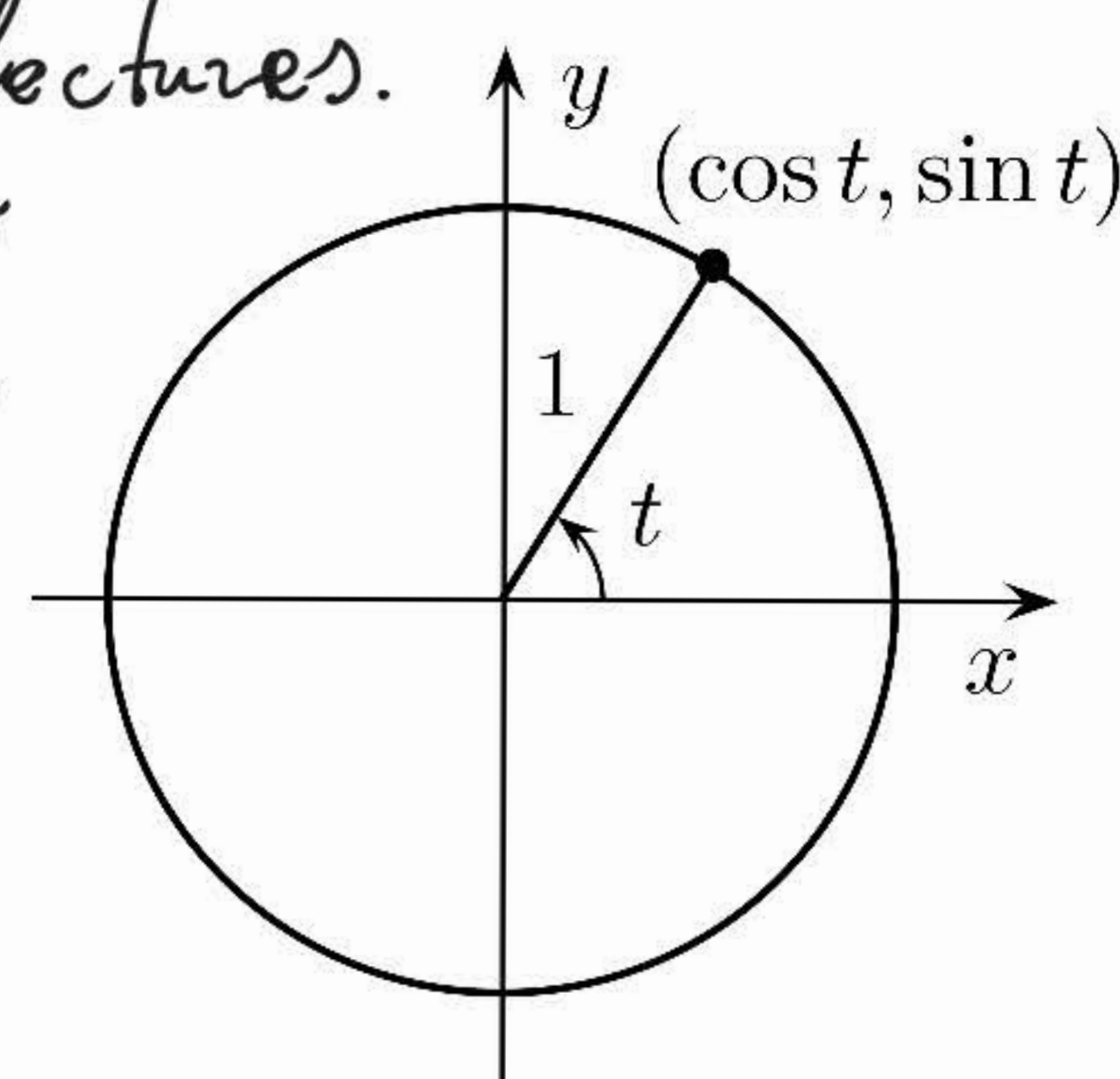


Question 2) $\sin \theta = \frac{b}{c}$; Assuming b is directional, its value can be at least -c and at most, +c. $\Rightarrow -1 \leq \sin \theta \leq +1$
Similar arguments follow for $\cos \theta$, $\tan \theta$, $\cot \theta$.

Question 3) $a^2 + b^2 = c^2 \rightarrow \frac{a^2}{c^2} + \frac{b^2}{c^2} = \frac{c^2}{c^2} = 1$
 $= \sin^2 \theta + \cos^2 \theta$ QED.

Question 4) Recall the unit circle in the lectures.

From symmetry arguments, $\begin{cases} \sin(-\theta) = -\sin \theta \\ \cos(-\theta) = \cos \theta \\ \dots \end{cases}$



Question 5)

$$\sin(\alpha + \beta) = \frac{AD}{OD} = \frac{DE}{OD} + \frac{EA}{OD} \quad \textcircled{I}$$

$$OD = \frac{DC}{\sin \beta}; \quad DE = (\cos \alpha) DC$$

$$\Rightarrow \frac{DE}{OD} = \cos \alpha \sin \beta \quad \textcircled{II}$$

$$EA = BC = \frac{OC}{\sin \alpha}; \quad OD = \frac{OC}{\cos \beta} \Rightarrow \frac{EA}{OD} = \sin \alpha \cos \beta \quad \textcircled{III}$$

$$I, II, III \Rightarrow \sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta \quad \textcircled{IV}$$

Using IV and replacing β with $-\beta$, it follows that,

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \sin\left(\frac{\pi}{2} - \alpha - \beta\right) = \sin\left(\frac{\pi}{2} - \alpha\right) \cos \beta - \cos\left(\frac{\pi}{2} - \alpha\right) \sin \beta$$

$$\Rightarrow \cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\Rightarrow \cos(\alpha + (-\beta)) = \cos \alpha \cos(-\beta) - \sin \alpha \sin(-\beta)$$

$$\Rightarrow \cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

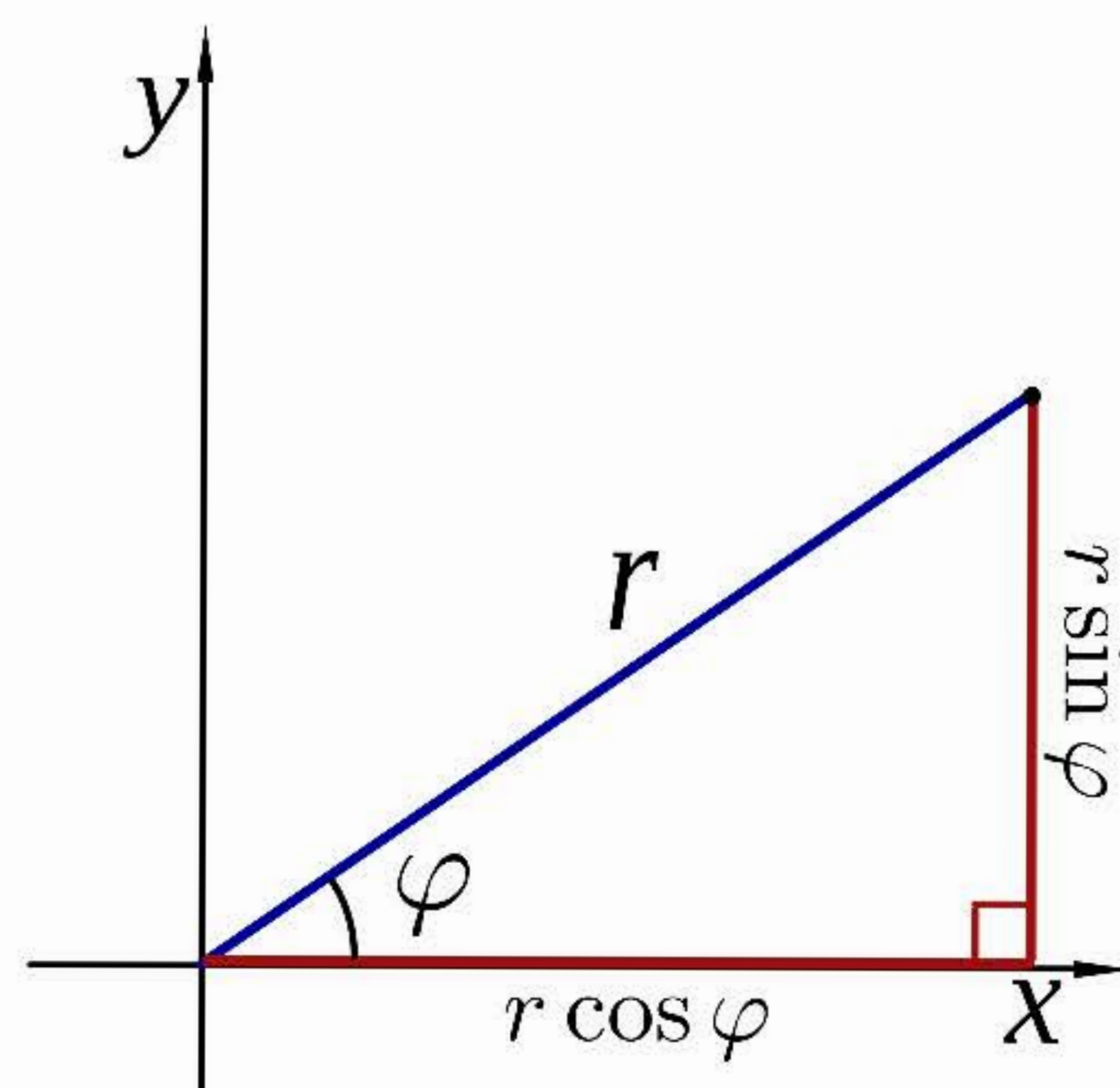
$$\tan(\alpha \pm \beta) = \frac{\sin(\alpha \pm \beta)}{\cos(\alpha \pm \beta)} = \frac{\sin \alpha \cos \beta \pm \cos \alpha \sin \beta}{\cos \alpha \cos \beta \mp \sin \alpha \sin \beta} \times \frac{\cos \alpha \cos \beta}{\cos \alpha \cos \beta}$$

$$= \frac{\tan \alpha \pm \tan \beta}{1 \mp \tan \alpha \tan \beta} \quad \text{QED.}$$

Question 6) $\begin{cases} x = r \cos \varphi \\ y = r \sin \varphi \end{cases}$

$$\Rightarrow \begin{cases} r^2(\sin^2 \varphi + \cos^2 \varphi) = x^2 + y^2 \\ \frac{y}{x} = \tan \varphi \end{cases}$$

$$\Rightarrow \begin{cases} r = \sqrt{x^2 + y^2} \\ \varphi = \arctan\left(\frac{y}{x}\right) \end{cases}$$



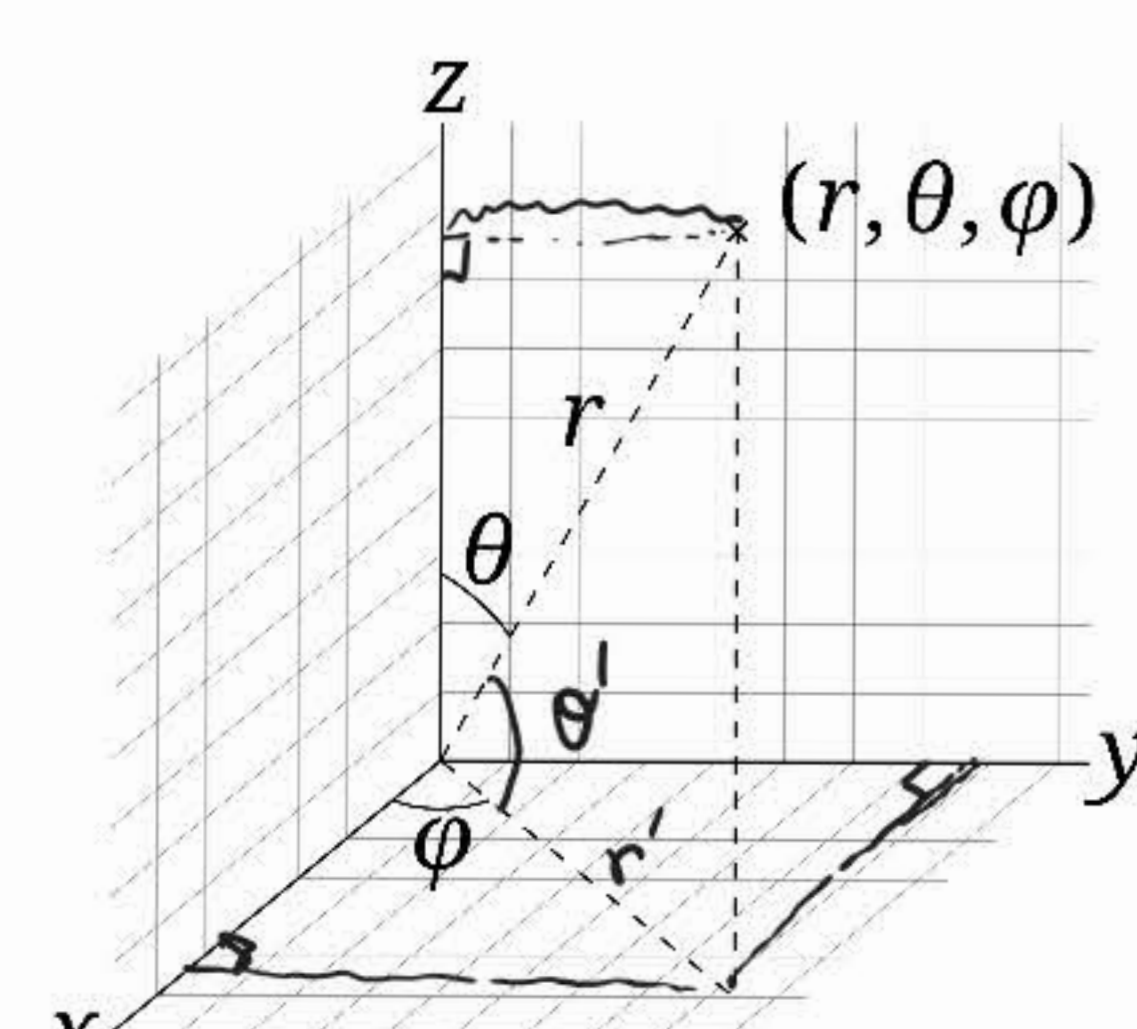
Question 7) $\begin{cases} x = r \cos \theta \cos \varphi \\ y = r \cos \theta \sin \varphi \\ z = r \sin \theta \end{cases}$

$$\Rightarrow r^2(\sin^2 \theta + \cos^2 \theta(\sin^2 \varphi + \cos^2 \varphi)) = x^2 + y^2 + z^2$$

$$\Rightarrow r^2 = x^2 + y^2 + z^2 \Rightarrow r = \sqrt{x^2 + y^2 + z^2}$$

$$\frac{y}{x} = \tan \varphi \Rightarrow \varphi = \arctan\left(\frac{y}{x}\right)$$

$$z = \sin \theta \sqrt{x^2 + y^2 + z^2} \Rightarrow \theta = \arcsin\left(\frac{z}{\sqrt{x^2 + y^2 + z^2}}\right)$$



Problem 8) $\begin{cases} x = \rho \cos \varphi \\ y = \rho \sin \varphi \\ z = z \end{cases} \Rightarrow \begin{cases} x^2 + y^2 = \rho^2 \\ \frac{y}{x} = \tan \varphi \end{cases}$

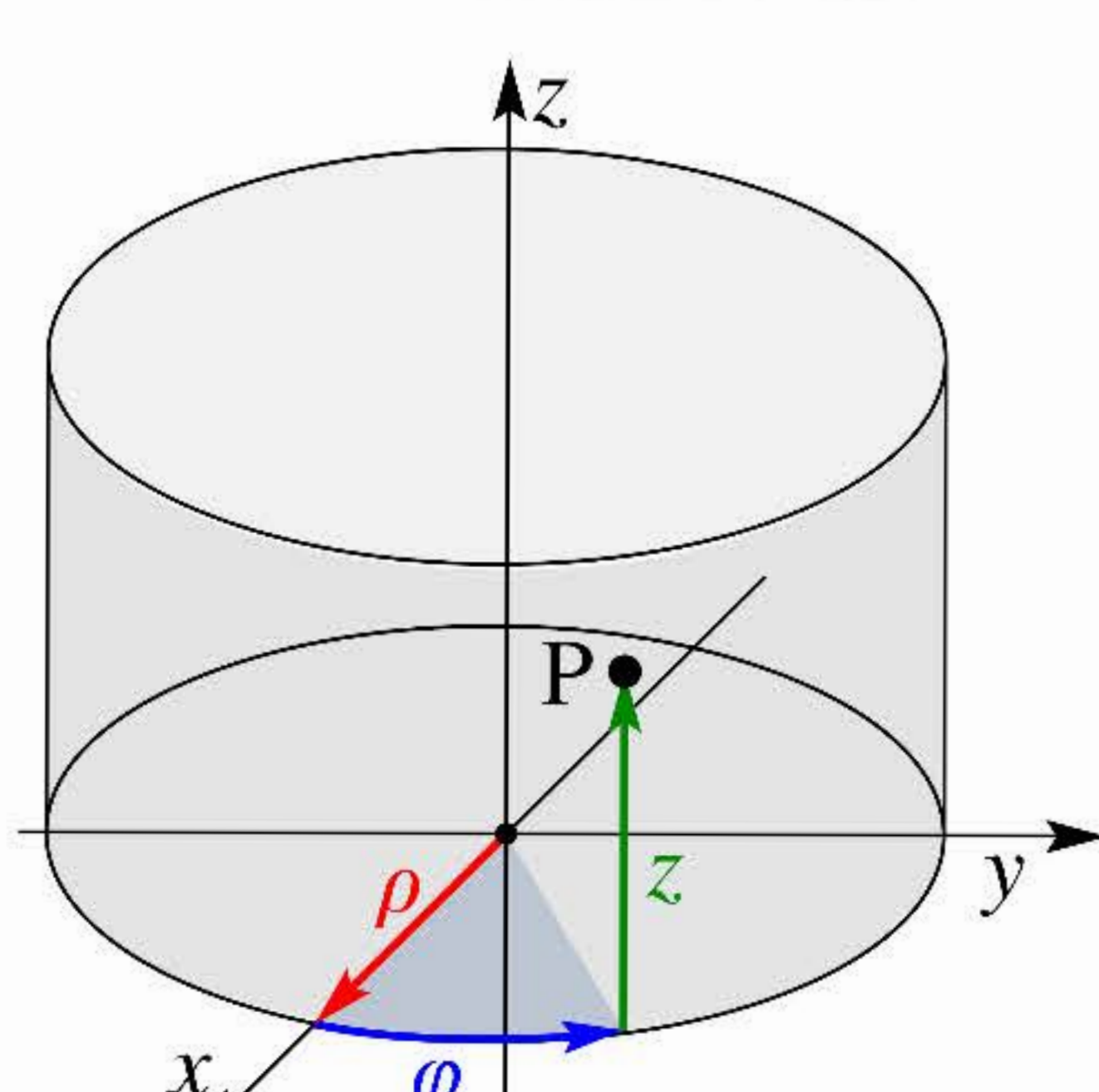
$$\Rightarrow \begin{cases} \rho = \sqrt{x^2 + y^2} \\ \varphi = \arctan\left(\frac{y}{x}\right) \\ z = z \end{cases}$$

Cylindrical to Cartesian

For Spherical, replace the Cartesian coordinates above with their corresponding spherical counterparts.

$$\begin{cases} x = r \cos \theta \cos \varphi \\ y = r \cos \theta \sin \varphi \\ z = r \sin \theta \end{cases} \Rightarrow \begin{cases} \rho = r \sqrt{\cos^2 \theta \cos^2 \varphi + \cos^2 \theta \sin^2 \varphi} = r \cos \theta \\ \varphi = \arctan\left(\frac{r \cos \theta \sin \varphi}{r \cos \theta \cos \varphi}\right) = \varphi \\ z = r \sin \theta \end{cases}$$

For the reverse, note $\begin{cases} \frac{z}{\rho} = \tan \theta \\ \varphi = \varphi \\ z = r \sin \theta \end{cases} \Rightarrow \begin{cases} \theta = \arctan\left(\frac{z}{\rho}\right) \\ \varphi = \varphi \\ r = \sqrt{\rho^2 + z^2} \end{cases}$



Problem 9) All three are valid, as long as one-to-one mapping is ensured.